

VPA ICT — IGCSE ICT Year 9 Semester 1: Theory Foundations

IGCSE ICT (0417) — Year 9 Semester 1: Theory Foundations

Syllabus: CIE IGCSE ICT (0417) | Exam: Paper 1 — Theory (2 hours, 100 marks)

Duration: 18 weeks | Lesson Length: 40 minutes

Prerequisites: Completion of Grades 6–8 ICT program or equivalent

Paper 1 Syllabus Coverage This Semester

Topic	Syllabus Ref	Weeks	Marks Weight
Types and Components of Computer Systems	1.1 – 1.3	1 – 2	~8%
Input and Output Devices	2.1 – 2.3	3 – 4	~10%
Storage Devices and Media	3.1 – 3.3	5 – 6	~8%
Computer Networks	4.1 – 4.5	7 – 9	~12%
Data Types	5.1 – 5.3	10 – 11	~8%
Effects of Using ICT	6.1 – 6.4	12 – 13	~10%
Safety and Security	9.1 – 9.3	14 – 15	~10%
Systems Analysis and Design	8.1 – 8.5	16 – 17	~8%
Revision & Assessment	All	18	—

Assessment Structure

Paper 1 Theory (2 hours, 100 marks):

Section A: Short-answer questions (approx. 30 marks)

Section B: Structured questions requiring longer responses (approx. 70 marks)

Question Types:

Knowledge recall (define, state, list)

Understanding (explain, describe, compare)

Application (suggest, discuss, evaluate)

Semester Learning Objectives

By the end of Semester 1, students will be able to:

- ☐ Describe the components of computer systems and their functions
- ☐ Compare input and output devices for different scenarios
- ☐ Evaluate storage solutions based on capacity, speed, portability, and durability
- ☐ Explain network types, topologies, and protocols

- ☐ Identify and use appropriate data types in ICT contexts
- ☐ Analyze the social, economic, and environmental effects of ICT
- ☐ Describe threats to data security and appropriate countermeasures
- ☐ Outline the systems development life cycle

Bilingual Support for IGCSE

All theory topics include:

Bilingual keyword lists (EN/中文/Pinyin) for every topic

Simplified Chinese summaries of complex concepts

Past paper questions with annotated mark schemes

Sentence frames for "explain" and "discuss" questions

Differentiation for IGCSE Theory

Level	Support	Extension
Bronze	Bilingual notes, cloze passages, matching exercises	—
Silver	Structured note templates, guided past paper practice	Additional application questions
Gold	Independent research, peer teaching	Extension reading on emerging technologies; explore A-Level topics

Resources

CIE IGCSE ICT (0417) syllabus (latest version)

CIE past papers (2018 – 2024)

"IGCSE ICT" by Graham Brown and David Watson (Hodder Education)

Cambridge Elevate digital resources

Bilingual glossary (see Supporting Documents)

Quizlet flashcard sets for each topic